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FOC Transformation for Single Open-Phase Faults in the Five-Phase Open-End Winding Topology

Matthew Priestley , Mohammad Farshadnia , and John Edward Fletcher , *Senior Member, IEEE*

Abstract—Many critical five-phase permanent-magnet synchronous machine drive systems require operation after a single open-phase fault. The single dc supply open-end winding inverter topology allows for the injection of four unbalanced phase currents to better optimize the post-fault machine torque after this fault. However, all previously published field-oriented control schemes send varying current references to the proportional–integral (PI) current controllers when controlling four unbalanced phase currents. This makes the design of the PI controllers considerably difficult as each machine operating point will require a different gain to accurately track the varying reference with minimum ripple. Consequently, this paper proposes a new postfault transformation matrix that obtains constant current references for any given set of four unbalanced sinusoidal phase currents. This facilitates easy design of the PI current controllers as the steady-state error can be reduced to zero in a user-defined response time for all operating points. This current control technique is the first to be designed specifically for the case of an open-phase fault to a five-phase open-end winding system.

Index Terms—Fault tolerant control, motor drives, permanent-magnet machines, proportional–integral (PI) control.

NOMENCLATURE

1, 2, 3, 4	Nonfaulted (healthy) phase number.
$C_{\text{rotational}}$	Conventional rotational transformation.
$d_f, q_f, t_{1,2f}$	New postfault d-, q-, tangent 1 or 2 axis.
i	Machine winding current (A).
I_m	Amplitude of stator current (A).
$i_{x,f}$	postfault phase ‘x’ current (A).
L	Machine winding inductance (H).
$L_{d,f}$	Proposed postfault transformed average inductance (H).

$L_{d,f-1}$

$L_{x,y}$

p

R

t_1-t_4, n_1-n_4, u, v

$T_e(t)$

$T_{f,4\phi}(\theta)$

T_{stat}

v

α, β

δ_f

λ

λ_{PM}

λ_{PMx}

ω

Proposed postfault transformed instantaneous inductance (H).

Phase inductance of phase winding ‘x’ from phase winding ‘y’ (H).

Number of machine poles.

Resistance of phase motor winding (Ω).

Constant variables in $T_{f,4\phi}(\theta)$.

Output machine alignment torque (N·m).

Proposed postfault FOC transformation.

Proposed stationary transformation matrix.

Machine terminal voltage (V).

Phase-shift angle in postfault current (rad).

Phase-shift angle of faulted phase (rad).

Machine flux-linkage (Wb-t).

Motor rotor magnet flux-linkage (Wb-t).

Motor rotor magnet flux-linkage ‘x’ harmonic order magnitude (Wb-t).

Electrical speed of the motor (rad/s).

I. INTRODUCTION

THERE is a range of safety critical applications that utilize electric machines and do not tolerate any disturbance to operation. An electric drive system that can continue to operate after a fault has occurred is valuable for these applications. The permanent-magnet synchronous machine (PMSM) is being considered for many of these critical applications including wind turbine generation [1], naval propulsion systems [2], electric vehicle traction drives [3] and [4], and aircraft control systems [5]. These PMSM systems require a control technique that allows for postfault operation.

There are many postfault optimization techniques for a single open-phase fault in a PMSM system. These techniques optimize the remaining nonfaulted phase currents to self-heal the magnetic field that is affected by the fault. The method in [6] waveshapes the two nonfaulted phase currents in a three-phase star-connected PMSM system. This avoids parts of the machine rotational cycle experiencing opposite torque polarity. However, the limited degrees of freedom in this system cause high harmonic current solutions and large torque pulsation [7].

The five-phase star-connected PMSM offers more degrees of freedom for postfault optimization. Five-phase optimization strategies are presented in [8] and [9], which produce high post-fault current harmonic solutions. Therefore, only the first and

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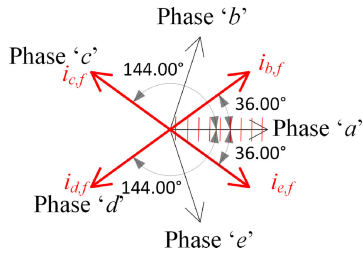


Fig. 1. Postfault current phasors for five-phase star-connected system during single open-phase fault.

third harmonic back-EMF and current components are included in [10]–[12], and this method is generalized for n -phases in [13]. The five-phase optimization techniques in [14] and [15] also consider the machine stator winding temperature and flux-weakening conditions respectively. However, these optimization strategies need a current control algorithm to control the four balanced phase current solutions.

One possible method of controlling these currents is to use an adapted field control (FOC) scheme. This control architecture requires a postfault transformation that is specifically designed for the desired postfault phase currents. The transformation in [16] and [17] is designed to send constant current references to the PI current controllers when injecting the sinusoidal phase currents in Fig. 1. Note that obtaining constant input references for the PI controllers is important as it simplifies the controller design process by allowing the usage of standard controller design techniques [16]. Another transformation in [18] achieves decoupled control of four balanced phase current solutions with fundamental and third harmonic components. However, the generated PI current controller references are no longer constant, which makes designing these controllers more difficult. Also, adding third harmonic components makes a tradeoff between reducing the maximum torque per ampere and reducing the torque pulsation.

When injecting sinusoidal phase currents, the postfault current phasors are constrained to be symmetrical and of equal amplitude (Fig. 1). This minimizes torque pulsation and avoids derating some of the phase currents, which avoids a reduction of the average torque [10]. However, these postfault current phasors produce considerable torque pulsation as only limited optimization is allowed. This torque pulsation can be reduced by removing the null zero-sequence constraint, which requires using another winding configuration.

There are some reported postfault optimization techniques that consider different five-phase winding configurations. The optimization methods in [19]–[21] consider the star, pentagon, and pentacle configurations. Another possible configuration that can remove the zero zero-sequence constraint is the single dc supply open-end winding configuration in Fig. 2. This topology allows for the injection of optimized unbalanced phase currents (i.e., currents with zero-sequence components). The additional zero-sequence component allows for further optimization of the machine output torque during a single open-phase fault [22]. However, this topology requires an extra PI current controller to control the zero-sequence current [23] and requires either PWM or the modulation technique in [24]. The technique in

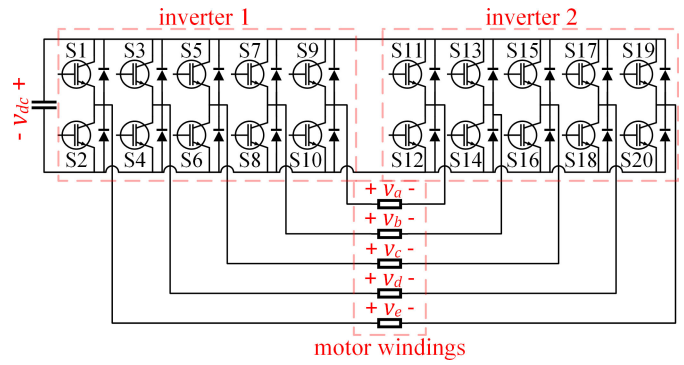


Fig. 2. Five-phase single dc supply two-level open-end winding topology circuit diagram.

[25] injects zero-sequence current to ensure all postfault phase currents are the same polarity for the entire machine rotational cycle. This means that the system can avoid isolating a phase during a diode or switch failure. However, the average torque is considerably de-rated with this approach. The technique in [26] proposes an online method for detecting an interturn short-circuited winding and minimizing the subsequent torque ripple.

At present, there are no current control strategies that have been designed specifically for implementing adapted FOC to a five-phase open-end winding PMSM system with an open-phase fault. All previous FOC techniques have transformations that produce varying current references for the PI current controllers when controlling four unbalanced phase currents. These varying references make the controllers difficult to tune for all operating points. Specifically, each torque and speed operating point will require a different gain to accurately track the varying reference with minimum ripple. Obtaining constant input references for the PI controllers means that the steady-state error can be reduced to zero within a user-defined response time. This ensures that a single set of gains can be used for all operating points and ensures that controllers can be easily designed to obtain the desired torque response.

Therefore, this paper presents a novel method for developing a postfault four-dimensional FOC transformation using the linear equations (2), (14), (15), (19)–(24). This transformation obtains constant current references for a given set of four unbalanced or balanced sinusoidal currents. The proposed transformation facilitates the easy design of the PI current controllers in an adapted FOC architecture (Fig. 3). This current control technique is the first to be designed specifically for the case of an open-phase fault to a five-phase open-end winding system.

II. NOVEL POSTFAULT FOUR-PHASE TRANSFORMATION

In this section, the technique for developing the proposed postfault four-phase transformation is presented for a given set of four unbalanced sinusoidal currents. This method utilizes simple linear equations to ensure that minimal computation time is required to generate the transformation real-time. These postfault current phasors must be symmetrical about the faulted phase to minimize torque pulsation [10]. Also, the amplitude of the postfault phase currents must be equal to avoid derating some of the phase currents and hence reducing the average

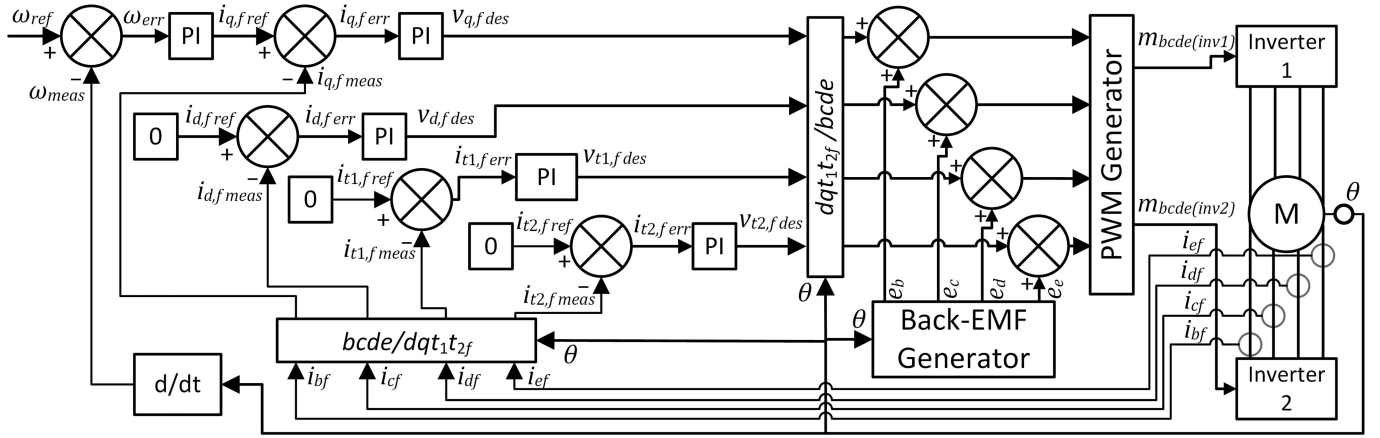


Fig. 3. Architecture of the postfault flux-oriented control strategy for a single open-phase fault to a five-phase open-end winding PMSM system.

torque [10]. Consequently, the postfault currents for a single open-phase fault are shown in (1) and the proposed postfault transformation is outlined in (2) shown at the bottom of this page.

$$\begin{bmatrix} i_{1,f} \\ i_{2,f} \\ i_{3,f} \\ i_{4,f} \end{bmatrix} = \begin{bmatrix} I_m \sin(\omega t - \delta_f - \alpha) \\ I_m \sin(\omega t - \delta_f - \beta) \\ I_m \sin(\omega t - \delta_f + \beta) \\ I_m \sin(\omega t - \delta_f - \alpha) \end{bmatrix} \quad (1)$$

Note that δ_f is the faulted phase's electrical phase-shift angle. For example, a fault in phase "a" means that δ_f is zero and $i_{1,f}$, $i_{2,f}$, $i_{3,f}$, $i_{4,f}$ equal $i_{b,f}$, $i_{c,f}$, $i_{d,f}$, $i_{e,f}$, respectively. Whereas, a fault to phase "c" means that δ_f is 144° and $i_{1,f}$, $i_{2,f}$, $i_{3,f}$, $i_{4,f}$ equal $i_{a,f}$, $i_{b,f}$, $i_{d,f}$, $i_{e,f}$, respectively. Note that the remainder of this manuscript assumes a fault in phase "a" for simplicity and hence δ_f is assumed to be zero. The two phase-shift angles α and β are determined using the optimization algorithm detailed in [27]. This algorithm first generates a set of initial phase currents over one electrical cycle. These currents and the offline recorded flux-linkages are then used to calculate a theoretical instantaneous torque array for one electrical cycle. This torque array is then used in an objective function to optimize both the postfault torque ripple and average torque per ampere. A differential evolution algorithm is then used to select a new set of postfault phase currents, in the form (1), which better comply with the objective function. This process is performed offline for a certain number of iterations until the phase currents, α and β , which give the desired torque characteristics are obtained. Note that this process is outlined in detail in [27].

The proposed postfault four-phase transformation matrix, (2) shown at the bottom of this page, is derived using a new stationary transformation matrix (3). This stationary transformation

has been adapted from the transformation in [16] to include the phase-shift angles from (1) in the first two rows. The new matrix, (3), has also been updated with unknown constants in the last two rows. These constants are later solved to ensure that the new postfault t_{1f} - and t_{2f} -axes possess zero input references for the PI current controllers, when controlling the postfault phase currents in (1). The variables u and v are later solved to ensure that the new postfault stationary axes from the first two rows transcribe a circular locus when controlling the postfault phase currents in (1). This allows the standard rotational transformation, (4), to be used to obtain the final proposed postfault transformation, (2), using (5). The four postfault transformed axes, from (2), are denoted as the d_f -axis, q_f -axis, t_{1f} -axis, and t_{2f} -axis. Note that these axes are not parallel or equal to the standard d -axis, q -axis, z -axis, and 0 -axis.

$$T_{\text{stat}} = \begin{bmatrix} u \cos(\alpha) & v \cos(\beta) & v \cos(-\beta) & u \cos(-\alpha) \\ u \sin(\alpha) & v \sin(\beta) & v \sin(-\beta) & u \sin(-\alpha) \\ t_1 & t_2 & t_3 & t_4 \\ n_1 & n_2 & n_2 & n_1 \end{bmatrix} \quad (3)$$

$$C_{\text{rotational}} = \begin{bmatrix} \cos(\omega t) & \sin(\omega t) & 0 & 0 \\ -\sin(\omega t) & \cos(\omega t) & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \quad (4)$$

$$T_{4\phi}(\theta) = C_{\text{rotational}} T_{\text{stat}}. \quad (5)$$

The t_{1f} - and t_{2f} -axis rows in the proposed transformation, (2), only contain constant terms. The postfault currents in (1) are symmetrical about the faulted phase. Therefore, the magnitudes (v and u) and phase-angles of the d_f -, q_f -, and t_{2f} -axis components in the proposed transformation matrix also have the same symmetry. The solving of the variables in (2) involves

$$T_{4\phi}(\theta) = \begin{bmatrix} u \cos(\omega t - \delta_f - \alpha) & v \cos(\omega t - \delta_f - \beta) & v \cos(\omega t - \delta_f + \beta) & u \cos(\omega t - \delta_f + \alpha) \\ -u \sin(\omega t - \delta_f - \alpha) & -v \sin(\omega t - \delta_f - \beta) & -v \sin(\omega t - \delta_f + \beta) & -u \sin(\omega t - \delta_f + \alpha) \\ t_1 & t_2 & t_3 & t_4 \\ n_1 & n_2 & n_2 & n_1 \end{bmatrix} \quad (2)$$

establishing multiple constraints. The first constraint is that the postfault currents, (1), must give zero d_f -axis current (6). The second constraint is that the currents, (1), must give a value of I_m current in the q_f -axis (7). These two constraints are solved to obtain (8) and (9).

$$\begin{aligned} & u \cos(\omega t - \alpha) \sin(\omega t - \alpha) + v \cos(\omega t - \beta) \sin(\omega t - \beta) \\ & + u \cos(\omega t + \alpha) \sin(\omega t + \alpha) \\ & + v \cos(\omega t + \beta) \sin(\omega t + \beta) = 0 \end{aligned} \quad (6)$$

$$\begin{aligned} & u \sin(\omega t - \alpha) \sin(\omega t - \alpha) + v \sin(\omega t - \beta) \sin(\omega t - \beta) \\ & + u \sin(\omega t + \alpha) \sin(\omega t + \alpha) \\ & + v \sin(\omega t + \beta) \sin(\omega t + \beta) = 1 \end{aligned} \quad (7)$$

$$u \cos(2\alpha) + v \cos(2\beta) = 0 \quad (8)$$

$$u + v = 1. \quad (9)$$

The next constraint is that the currents, (1), must give zero t_{2f} -axis current (10). Also, the unit vector length of the t_{2f} -axis current component must be equal to one. These two constraints yield (11) and (12). In addition to this, the t_{1f} -axis must be orthogonal to the t_{2f} -axis (13), which yields (14) and (15).

$$\begin{aligned} & n_1 \cos(\omega t - \alpha) + in_1 \sin(\omega t - \alpha) + n_2 \cos(\omega t - \beta) \\ & + in_2 \sin(\omega t - \beta) + n_2 \cos(\omega t + \beta) + in_2 \sin(\omega t + \beta) \\ & + n_1 \cos(\omega t + \alpha) + in_1 \sin(\omega t + \alpha) = 0 \end{aligned} \quad (10)$$

$$n_1 \cos(\alpha) + n_2 \cos(\beta) = 0 \quad (11)$$

$$2n_1^2 + 2n_2^2 = 1 \quad (12)$$

$$n_1 t_1 + n_2 t_2 + n_2 t_3 + n_1 t_4 = 0 \quad (13)$$

$$t_4 = -t_1 \quad (14)$$

$$t_3 = -t_2. \quad (15)$$

The currents, (1), must also give zero t_{1f} -axis current (16). This yields the constraints (17) and (18). Therefore, (8), (9), (14), (15), (17), and (18) can be solved to obtain (19)–(24). The proposed four-phase postfault transformation matrix in (2) is defined using (14), (15), (19)–(24).

$$\begin{aligned} & t_1 \cos(\omega t - \alpha) + it_1 \sin(\omega t - \alpha) + t_2 \cos(\omega t - \beta) \\ & + it_2 \sin(\omega t - \beta) - t_2 \cos(\omega t + \beta) - it_2 \sin(\omega t + \beta) \\ & - t_1 \cos(\omega t + \alpha) - it_1 \sin(\omega t + \alpha) = 0 \end{aligned} \quad (16)$$

$$t_1 \sin(\alpha) + t_2 \sin(\beta) = 0 \quad (17)$$

$$2t_1^2 + 2t_2^2 = 1 \quad (18)$$

$$u = \frac{\cos(2\beta)}{\cos(2\beta) - \cos(2\alpha)} \quad (19)$$

$$v = \frac{\cos(2\alpha)}{\cos(2\alpha) - \cos(2\beta)} \quad (20)$$

$$n_1 = \sqrt{\frac{\cos^2(\beta)}{2(\cos^2(\alpha) + \cos^2(\beta))}} \quad (21)$$

$$n_2 = \sqrt{\frac{\cos^2(\alpha)}{2(\cos^2(\alpha) + \cos^2(\beta))}} \quad (22)$$

$$t_1 = \sqrt{\frac{\sin^2(\beta)}{2(\sin^2(\alpha) + \sin^2(\beta))}} \quad (23)$$

$$t_2 = -\sqrt{\frac{\sin^2(\alpha)}{2(\sin^2(\alpha) + \sin^2(\beta))}}. \quad (24)$$

III. IMPLEMENTATION OF NEW TRANSFORMATION

This section outlines how the proposed four-dimensional (4-D) transformation is implemented in an adapted FOC architecture. The FOC strategy requires that the proposed transformation matrix has a valid inverse matrix (25) shown at the bottom of this page. Also, the transformed machine equations need to be determined to be able to accurately design the PI current controllers in the FOC scheme, where $N = 2t_2 u \sin(\alpha) - 2t_1 v \sin(\beta)$ and $T = 2n_2 u \cos(\alpha) - 2n_1 v \cos(\beta)$.

The postfault machine equations in phase domain form are (26). Applying the proposed transformation $T_{f,4\phi}(\theta)$ to these equations yields (27). The proposed transformation complies with the matrix property in (28), which simplifies the complex term in (27) to a simple cross-product in (29). The final term in (29), which contains the transformed back-EMF terms, possesses significant cross-coupling between the four transformed axes. This cross-coupling occurs because the proposed transformation phase-shift angles no longer equal the five-phase back-EMF harmonic series phase-shift angles. This is not desired as this significant cross-coupling causes high harmonic output and potential controller instability. Therefore, the machine phase back-EMFs are fed-forward in the adapted FOC architecture to avoid the PI current controllers compensating for the phase back-EMFs in the electrical machine equations.

$$v_{bcde} = Ri_{bcde} + \frac{d(L_{bcde} i_{bcde})}{dt} + e_{bcde} \quad (26)$$

$$\begin{aligned} v_{dq1t2,f} &= Ri_{dq1t2,f} + \frac{d(L_{dq1t2,f} i_{dq1t2,f})}{dt} \\ &+ T_{f,4\phi}(\theta) \frac{d}{dt} \left(T_{f,4\phi}(\theta)^{-1} \right) \\ &\times L_{dq1t2,f} i_{dq1t2,f} + T_{f,4\phi}(\theta) e_{bcde} \end{aligned} \quad (27)$$

$$T_{f,4\phi}(\theta)^{-1} = \begin{bmatrix} \cos(\omega t - \alpha) - \sin(\omega t - \alpha) - v \sin(\beta)/N & -v \cos(\beta)/T \\ \cos(\omega t - \beta) - \sin(\omega t - \beta) & u \sin(\alpha)/N & u \cos(\alpha)/T \\ \cos(\omega t + \beta) - \sin(\omega t + \beta) & -u \sin(\alpha)/N & u \cos(\alpha)/T \\ \cos(\omega t + \alpha) - \sin(\omega t + \alpha) & v \sin(\beta)/N & -v \cos(\beta)/T \end{bmatrix} \quad (25)$$

$$T_{f,4\phi}(\theta) \frac{d}{dt} \left(T_{f,4\phi}(\theta)^{-1} \right) X = \begin{bmatrix} 0 & -\omega & 0 & 0 \\ \omega & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} X = \omega \times X \quad (28)$$

$$v_{dqt1t2,f} = R i_{dqt1t2,f} + \frac{d(L_{dqt1t2,f} i_{dqt1t2,f})}{dt} + \omega \times L_{dqt1t2,f} i_{dqt1t2,f} + T_{f,4\phi}(\theta) e_{bcde}. \quad (29)$$

The transformed inductance matrix $L_{dqt1t2,f}$, which is calculated by (30), also possesses harmonic components in all the terms in this matrix (31). This is since the transformation phase-shift angles no longer equal the five-phase self- and mutual-inductance harmonic series phase-shift angles. These harmonic terms will also cause undesired cross-coupling between the transformed axes. Note that the self- and mutual-inductance terms depend on current magnitude, magnetic saturation, and rotor-position. This makes obtaining accurate terms in $L_{dqt1t2,f}$ for all operating points difficult. Consequently, the inductance harmonic terms cannot be accurately fed-forward.

$$L_{dqt1t2,f} = T_{4\phi}(\theta) \begin{bmatrix} L_{bb} & L_{bc} & L_{bd} & L_{be} \\ L_{bc} & L_{cc} & L_{cd} & L_{ce} \\ L_{bd} & L_{cd} & L_{dd} & L_{de} \\ L_{be} & L_{ce} & L_{de} & L_{ee} \end{bmatrix} T_{4\phi}(\theta)^{-1} \quad (30)$$

$$L_{dqt1t2,f} = \begin{bmatrix} L_{d,f,1} & L_{d,f,2} & L_{d,f,3} & L_{d,f,4} \\ L_{q,f,1} & L_{q,f,2} & L_{q,f,3} & L_{q,f,4} \\ L_{t1,f,1} & L_{t1,f,2} & L_{t1,f,3} & L_{t1,f,4} \\ L_{t2,f,1} & L_{t2,f,2} & L_{t2,f,3} & L_{t2,f,4} \end{bmatrix}. \quad (31)$$

These harmonic terms in $L_{dqt1t2,f}$ will contribute some cross-coupling even during steady-state conditions. Therefore, a fault-tolerant motor should be designed with a constant self-inductance and small mutual inductance to minimize this cross-coupling and allow for the assumption in (32). The electrical machine equations that the PI current controllers must compensate for are (33), if the back-EMF is fed-forward and the magnitude of the machine inductance harmonics is small.

$$L_{dqt1t2,f} \approx \begin{bmatrix} L_{d,f} & 0 & 0 & 0 \\ 0 & L_{q,f} & 0 & 0 \\ 0 & 0 & L_{t1,f} & 0 \\ 0 & 0 & 0 & L_{t2,f} \end{bmatrix} \quad (32)$$

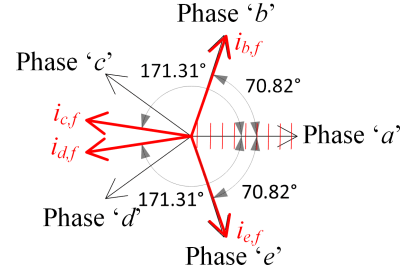


Fig. 4. Phasor diagram of a single open-phase fault with postfault optimized current phasors.

$$\begin{bmatrix} v_{d,f} \\ v_{q,f} \\ v_{t1,f} \\ v_{t2,f} \end{bmatrix} = R \begin{bmatrix} i_{d,f} \\ i_{q,f} \\ i_{t1,f} \\ i_{t2,f} \end{bmatrix} + \begin{bmatrix} L_{q,f} \frac{di_{d,f}}{dt} \\ [4pt] L_{q,f} \frac{di_{q,f}}{dt} \\ L_{t1,f} \frac{di_{t1,f}}{dt} \\ [4pt] L_{t2,f} \frac{di_{t2,f}}{dt} \end{bmatrix} + \begin{bmatrix} -\omega L_{q,f} i_{q,f} \\ [4pt] \omega L_{d,f} i_{d,f} \\ 0 \\ 0 \end{bmatrix} \quad (33)$$

$$T_e(t) = \frac{p}{2} \left[\begin{pmatrix} -L_{q,f} i_{q,f} + \lambda_{PMq,f} \\ L_{d,f} i_{d,f} + \lambda_{PMd,f} \\ 0 \\ 0 \end{pmatrix} + \frac{d}{d\omega t} \begin{pmatrix} \lambda_{PMd,f} \\ \lambda_{PMq,f} \\ \lambda_{PMt1,f} \\ \lambda_{PMt2,f} \end{pmatrix} \right] \times \left(T_{f,4\text{-phase}}(\theta)^{-1} \right)^T T_{f,4\text{-phase}}(\theta)^{-1} i_{dqt1t2,f}. \quad (34)$$

The new torque equation for the proposed transformation is dependent on the postfault currents and the permanent magnet (PM) flux-linkage terms (34). Hence, a generic simplified torque equation cannot be obtained, and a specific torque equation must be derived for the specific machine and fault combination. The proposed postfault FOC architecture is presented in Fig. 3.

IV. EXAMPLE POSTFAULT TRANSFORMATION

For the purposes of experimental testing, an example four-phase postfault transformation has been created in (35) shown at the bottom of this page, assuming a single open-phase fault in phase 'a'. A postfault optimization algorithm has been used to establish the postfault current angles $\alpha = 70.82^\circ$ and $\beta = 171.31^\circ$ to minimize torque ripple as shown in Fig. 4. Note that the machine specifications used in this optimization are presented in Table III in the Appendix. The transformed torque equation for the single open-phase fault with the optimized current angles is (36), where λ_{PM1} and λ_{PM3} are the PM flux-linkage fundamental and third harmonic amplitudes and

$$T_{f,4\text{phase}}(\theta) = \begin{bmatrix} 0.549 \cos(\omega t - 70.82^\circ) & 0.451 \cos(\omega t - 171.31^\circ) & 0.451 \cos(\omega t + 171.31^\circ) & 0.549 \cos(\omega t + 70.82^\circ) \\ -0.549 \sin(\omega t - 70.82^\circ) & -0.451 \sin(\omega t - 171.31^\circ) & -0.451 \sin(\omega t + 171.31^\circ) & -0.549 \sin(\omega t + 70.82^\circ) \\ 0.1117 & -0.6982 & 0.6982 & -0.1117 \\ 0.6710 & 0.2230 & 0.2230 & 0.6710 \end{bmatrix} \quad (35)$$

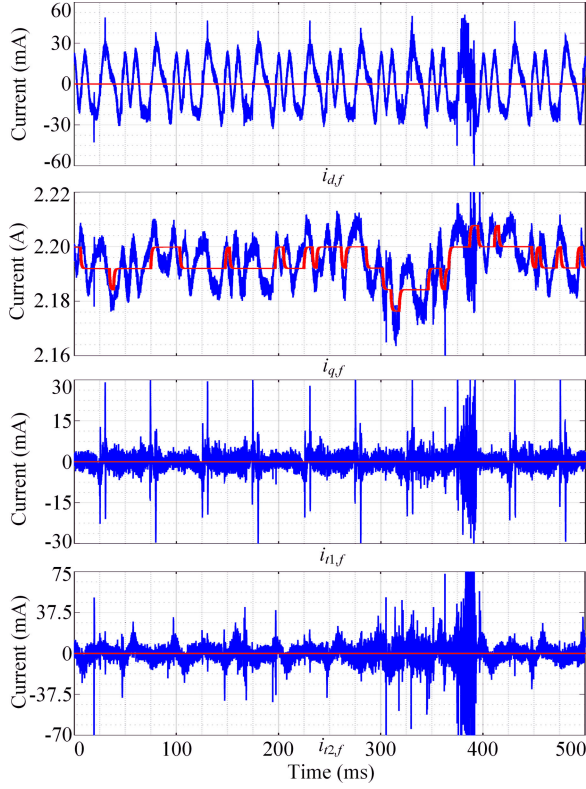


Fig. 5. Experimental transformed currents when operating with a rated current load torque at 300 rpm. Red line: reference current. Blue line: measured current.

$$i_{d,f} = i_{t1,f} = i_{t2,f} = 0.$$

$$T_e(t) = \frac{p}{2} i_{q,f} \left[\lambda_{PM1} \left(1.8886 - 0.0857 \sin \left(2\omega t - \frac{\pi}{2} \right) \right) - 3\lambda_{PM3} \left(0.9834 \sin \left(2\omega t + \frac{\pi}{2} \right) + 0.1601 \sin \left(4\omega t - \frac{\pi}{2} \right) \right) \right]. \quad (36)$$

V. EXPERIMENTAL RESULTS

An experimental test has been conducted to prove that the proposed postfault transformation can be used with PI current controllers in an FOC architecture to accurately control optimized postfault currents. This test has been conducted with a DS1006 dSPACE processor, which executes the control algorithm, and a DS5203 field programmable gate array, which generates the IGBT gating signals. Two five-phase two-level inverters have been connected in the open-end winding topology with a single 140 V dc bus and a prototype five-phase PMSM. Note that the specifications for this machine are presented in Table III of the Appendix. Note that phase ‘a’ has been disconnected from the inverter for this test. The proposed postfault current control technique with the example transformation matrix, (35), has been implemented with the machine operating at 300 and 1500 rpm. A load torque has then been applied to the machine such that the rated current is achieved. The results from these tests are shown in Figs. 5–11.

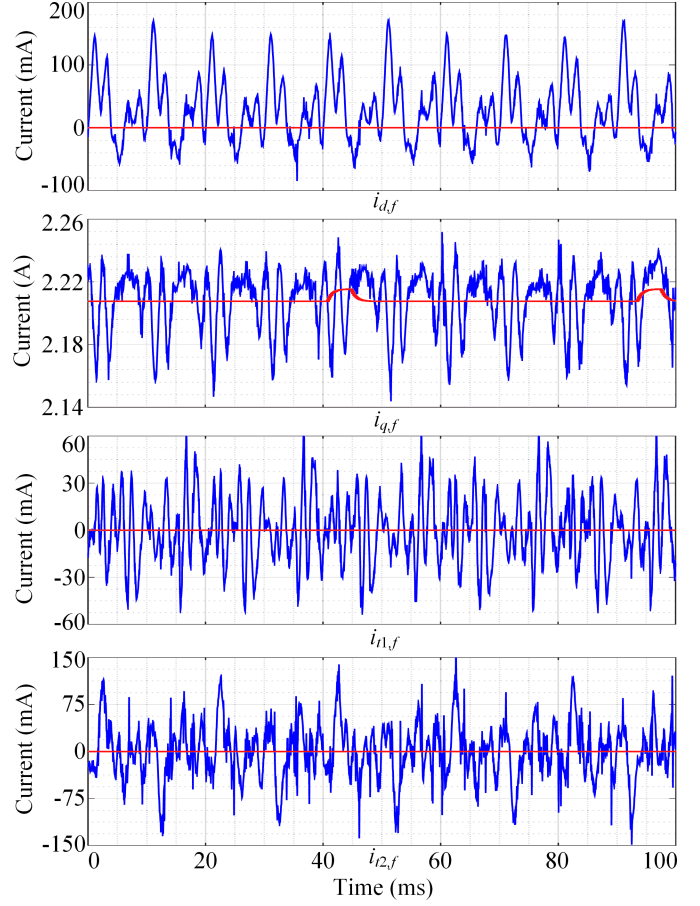


Fig. 6. Experimental transformed currents when operating with a rated current load torque at 1500 rpm. Red line: reference current. Blue line: measured current.

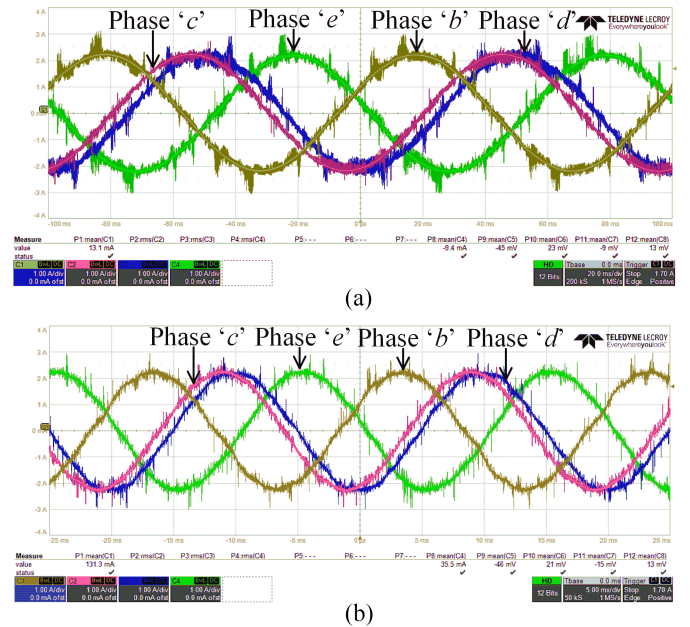


Fig. 7. Experimental measured phase currents. Current scale: 1 A/div. (a) Speed = 300 rpm (time scale: 20 ms/div). (b) Speed = 1500 rpm (time scale: 5 ms/div).

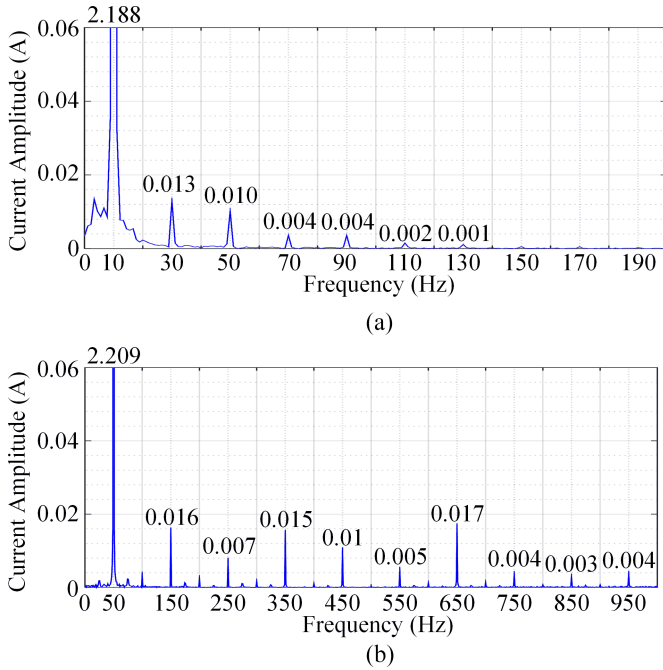


Fig. 8. Frequency spectra of measured phase currents. (a) Speed = 300 rpm. (b) Speed = 1500 rpm.

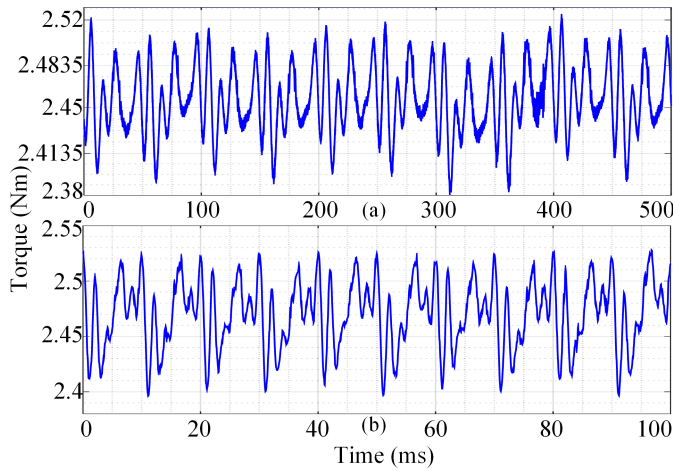


Fig. 9. Experimental developed torque when operating with a rated current load torque. (a) Speed = 300 rpm. (b) Speed = 1500 rpm.

At 300 rpm (Fig. 5), the measured $i_{q,f}$ waveform accurately tracks the $i_{q,f}$ reference waveform with approximately 50 mA of peak-to-peak ripple. Also, the $i_{d,f}$, $i_{t1,f}$, and $i_{t2,f}$ waveforms possess an average of 0 A with peak-to-peak current ripples of 60, 60, and 75 mA, respectively. At 1500 rpm (Fig. 6), the measured $i_{q,f}$ waveform accurately tracks the $i_{q,f}$ reference waveform but possesses an increased peak-to-peak ripple of approximately 100 mA. Also, the $i_{d,f}$, $i_{t1,f}$, and $i_{t2,f}$ waveforms have increased peak-to-peak ripples of approximately 240, 120, and 300 mA. Therefore, the current ripple increases as the speed increases.

This speed/current ripple relationship is due to the cross-coupling terms in the new postfault transformation machine equations. The transformed inductance matrix $L_{dq,t1t2,f}$ (31)

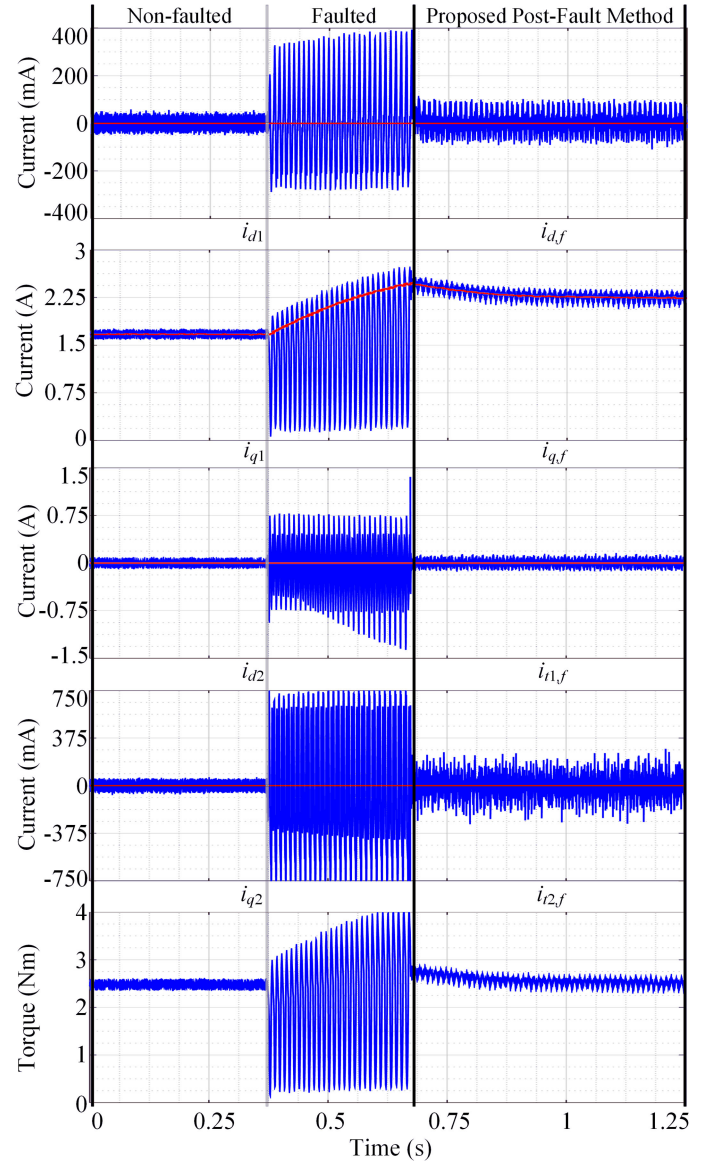


Fig. 10. Experimental transformed currents and developed torque, when operating with a rated current load torque at 1500 rpm, when transitioning from healthy (standard FOC) to faulty (standard FOC) and to postfault (proposed method).

possesses small harmonic components in each entry that will provide small amounts of cross-coupling between the axes. Also, the final terms of the transformed machine equations in (33) will contain $L_{d,f}$ and $L_{q,f}$. Therefore, the harmonics in these two transformed inductance terms will cause rotor-position variant and hence speed-variant cross-coupling. This cross-coupling will cause current harmonics (i.e., current ripple) that will increase as speed increases.

Also, there will be some current harmonic generation from small inaccuracies in the back-EMF feed-forwarding. This feed-forwarding is dependent on the measured PM flux-linkage being constant for all operating points. Note that small variations in the PM flux-linkage may occur across all operating points. The back-EMF feed-forwarding is also dependent on the speed calculation using the encoder position measurements. This

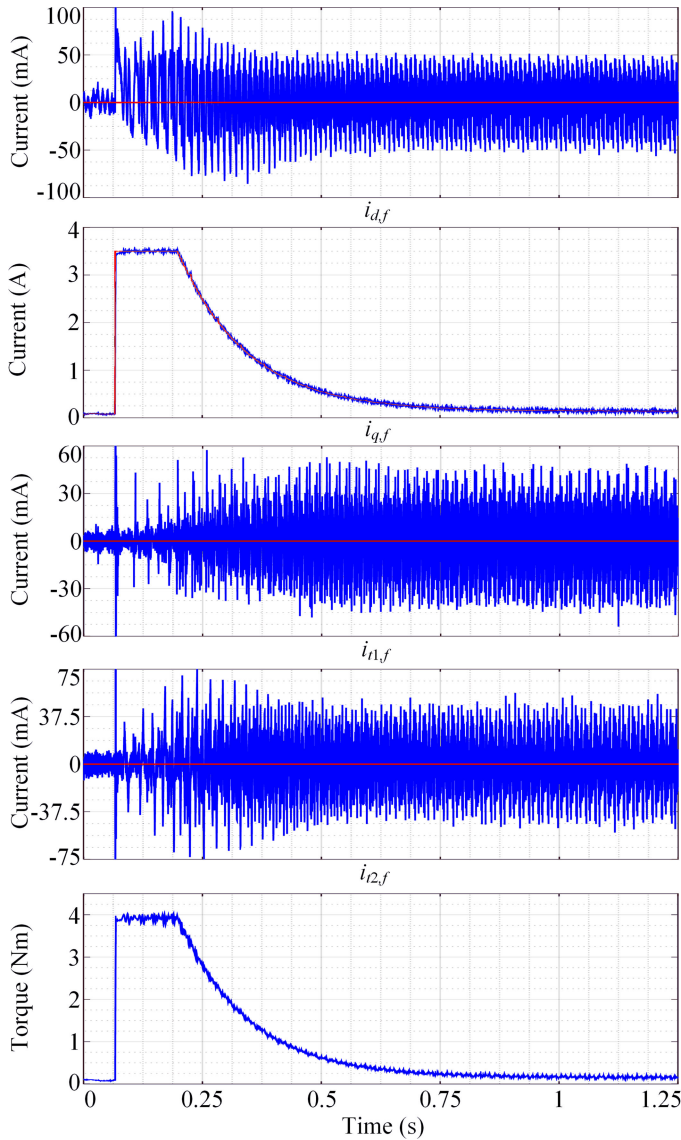


Fig. 11. Experimental transformed currents and developed torque during speed change from 300 to 1500 rpm.

calculation measures the change in rotor position across 80 control algorithm cycles, which gives a resolution of approximately 0.75 rpm when operating at 1500 rpm. These two small inaccuracies in the feed-forward back-EMF will produce some cross-coupling between axes that cause small speed-dependent current harmonics. The proposed strategy also has the limitation that the PI current controller gains will become dependent on any variations of the back-EMF waveforms, such as those due to the gradual aging of the permanent magnets. However, this limitation also occurs in the existing FOC methods and is the subject of future study.

The phase currents at 300 and 1500 rpm (Fig. 7) have minimal distortion and are sinusoidal in shape. This proves that the small magnitude current harmonics in the transformed axes responses produce negligible current harmonics in the phase domain. The frequency spectra of the phase currents (Fig. 8) also prove this as the harmonic content is insignificant when compared to the

TABLE I
PHASE-SHIFT ANGLES OF FUNDAMENTAL CURRENTS FROM FFT

Speed (rpm)	Phase 'b' angle	Phase 'c' angle	Phase 'd' angle	Phase 'e' angle
300	-73.3°	-173.8°	-191.2°	-291.6°
1500	-72.5°	-171.9°	-189.5°	-291.2°

fundamental amplitude. The phase-shift angles of the fundamental components in the frequency spectra (Table I) possess a small inaccuracy of approximately 2° (lagging) for each phase. This is likely due to the small distortion in the transformed current responses, encoder accuracy, and possibly a time delay somewhere in the system. This small phase-angle inaccuracy is shown to produce negligible torque ripple (Fig. 9).

The torque responses at both 300 and 1500 rpm (Fig. 9) have an average torque of 2.47 N·m with 0.14 N·m peak-to-peak ripple. Note that the rated torque of the machine is 3.2 N·m. Therefore, the proposed postfault transformation in an FOC architecture can produce approximately 77.2% of the average rated torque with a peak-to-peak torque ripple approximately equal to 5.7% of the average torque during a single open-phase fault at rated current.

The transformed currents and output torque have been recorded transitioning from healthy conditions to faulted conditions (Fig. 10). During this test, the machine is initially controlled at 1500 rpm using the standard five-phase transformation, with rated current load torque. An open-phase fault is then triggered on phase 'a', but the standard control method is still used for approximately 0.3 s. This demonstrates the effects of continuing to use the standard current control technique during an open-phase fault in an open-end winding system. The proposed postfault method is then implemented to demonstrate its effectiveness at regaining control of the machine. Under healthy conditions, the output torque ripple is approximately 0.109 N·m. During the faulted mode, when the standard control technique is still used, the system becomes unstable and the torque ripple continues to increase to significant levels. Continuing to operate in this mode will eventually damage the machine. During the postfault mode, when using the proposed method, the torque ripple is approximately 0.251 N·m while regaining control of the machine. This test proves the validity of the proposed method.

The transformed currents and output torque have also been recorded during a torque transient, which is obtained by changing the speed from 300 to 1500 rpm (Fig. 11). This test shows that the rise time and settling time of the torque response are the same as in the $i_{q,f}$ waveform. This validates that the proposed transformation facilitates the easy design of the PI current controllers. Note that this is achieved by sending constant current references to the PI current controllers.

Therefore, the results prove that the proposed postfault transformation can be used with PI current controllers and a back-EMF feed-forward controller to accurately control four unbalanced postfault phase currents with minimal harmonics or distortion in the open-end winding topology. The only limitation of this method is that the back-EMF must be recorded offline to feed-forward these waveforms. This is not a major disadvantage as measuring these waveforms is simple.

VI. CONCLUSION

A novel online method for developing a four-dimensional postfault transformation using simple linear equations, (2), (14), (15), (19)–(24), was detailed in this paper. This transformation obtains constant current references for any set of four unbalanced or balanced sinusoidal phase currents. The proposed transformation facilitates the easy design of the PI current controllers in an adapted FOC architecture (Fig. 3). This method was verified using an example transformation, (35), in an experimental setup with an open-phase fault to phase ‘a’. The results from this test proved that the proposed control technique can accurately control the unbalanced postfault sinusoidal currents with negligible distortion and harmonics. The results also showed that the proposed technique can produce approximately 77.2% of the average rated torque with a peak-to-peak torque ripple approximately equal to 5.7% of the average torque during a single open-phase fault. The results validate that the proposed transformation method is the easiest option for obtaining user-defined torque responses for all machine operating points.

APPENDIX

TABLE II
SYSTEM PARAMETERS FOR EXPERIMENTAL TESTS

Parameter	Value
DC bus voltage	140 V
Control algorithm processor model	dSPACE DS1006
IGBT gating signal generator	DS5203 FPGA
Rated peak motor current	2.25 A
Constant load torque	Configured to draw rated current
Motor speed reference	300 rpm and 1500 rpm
Motor rated speed	50 Hz

TABLE III

FIVE-PHASE PMSM BACK-EMF AND INDUCTANCE HARMONICS AT 50 Hz

Frequency (Hz)	Back-EMF voltage (V)	Self-inductance harmonics (mH)	Mutual inductance harmonics (mH)	
			Adj	Non-Adj
dc	0	24.66	7.86	2.57
50	93.16	0	0	0
100	0	2.91	1.34	0.07
150	8.34	0	0	0
200	0	1.62	0.45	0.52
250	1.16	0	0	0
300	0	0.21	0.22	0.10
350	0.4	0	0	0
400	0	0.03	0.02	0.02
450	1.64	0	0	0
500	0	0.09	0.07	0.01
550	2.35	0	0	0

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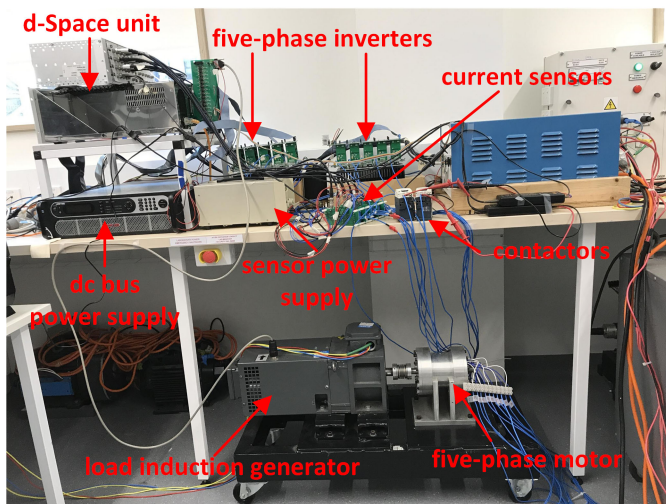
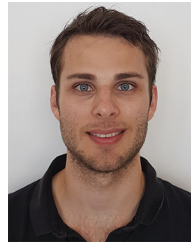


Fig. 12. Experimental setup used to obtain results in this paper.

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